



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Vikas Parmar
June 14, 2026



Vikas Parmar

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

This project predicts SpaceX Falcon 9 first-stage landing success using data science methods across the full workflow: API data collection, web scraping, data wrangling, SQL and visual EDA, Folium mapping, Plotly Dash analytics, and machine learning classification.

Vikas Parmar

Key result: Decision Tree

Introduction

- Project background and context
- Problems you want to find answers

SpaceX advertises Falcon 9 launches at about \$62M versus \$165M+ from other providers. Reusing the first stage drives much of that savings.

Problem: For a given launch (payload, orbit, site, booster version), will the Falcon 9 first stage land successfully?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data collection: SpaceX REST
API and Wikipedia scraping
Data wrangling: Missing
values, outcome mapping,
Class label
EDA: SQL queries +
Matplotlib/Seaborn charts
Interactive analytics: Folium
maps + Plotly Dash

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose

Place your flowchart of SpaceX API calls here

Flowchart:
SpaceX API -> requests.get()
-> json_normalize() -> Filter
Falcon 9 -> Fill missing
PayloadMass ->
dataset_part_1.csv

Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose

Place your flowchart of web scraping here

Flowchart:
Wikipedia page ->
requests.get() ->
BeautifulSoup -> parse
HTML table -> Pandas
DataFrame ->
spacex_web_scraped.csv

Data Wrangling

- Describe how data were processed
- You need to present your data wrangling process using key phrases and flowcharts
- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose

Flowchart:
dataset_part_1.csv -> EDA
summaries -> map Outcome
to Class (0/1) ->
dataset_part_2.csv +
spacex_launch_dash.csv

EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

Charts plotted:

- Flight Number vs Launch Site
- Payload vs Launch Site
- Success Rate vs Orbit Type
- Flight Number vs Orbit Type
- Payload vs Orbit Type
- Launch Success Yearly Trend

Vikas Parmar

GitHub:

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
- Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose

SQL queries performed:
SELECT DISTINCT launch
sites; CCA launch sites; NASA
CRS payload sum; F9 v1.1
average payload; first ground
landing date; drone ship
payload range;
success/failure counts; max
payload boosters; 2015
failures; ranked outcomes.

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
- Explain why you added those objects
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

Folium map objects:

- Markers for all launch sites
- Color-coded success/failure launch markers
- Proximity lines to coastline, highway, railway, and city

GitHub:

Vikas Parmar
[https://github.com/vikas-parmar/applied-data-science-](https://github.com/vikas-parmar/applied-data-science)

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose

Plotly Dash dashboard:

- Launch site dropdown
- Success pie chart
- Payload range slider
- Payload vs launch outcome scatter plot

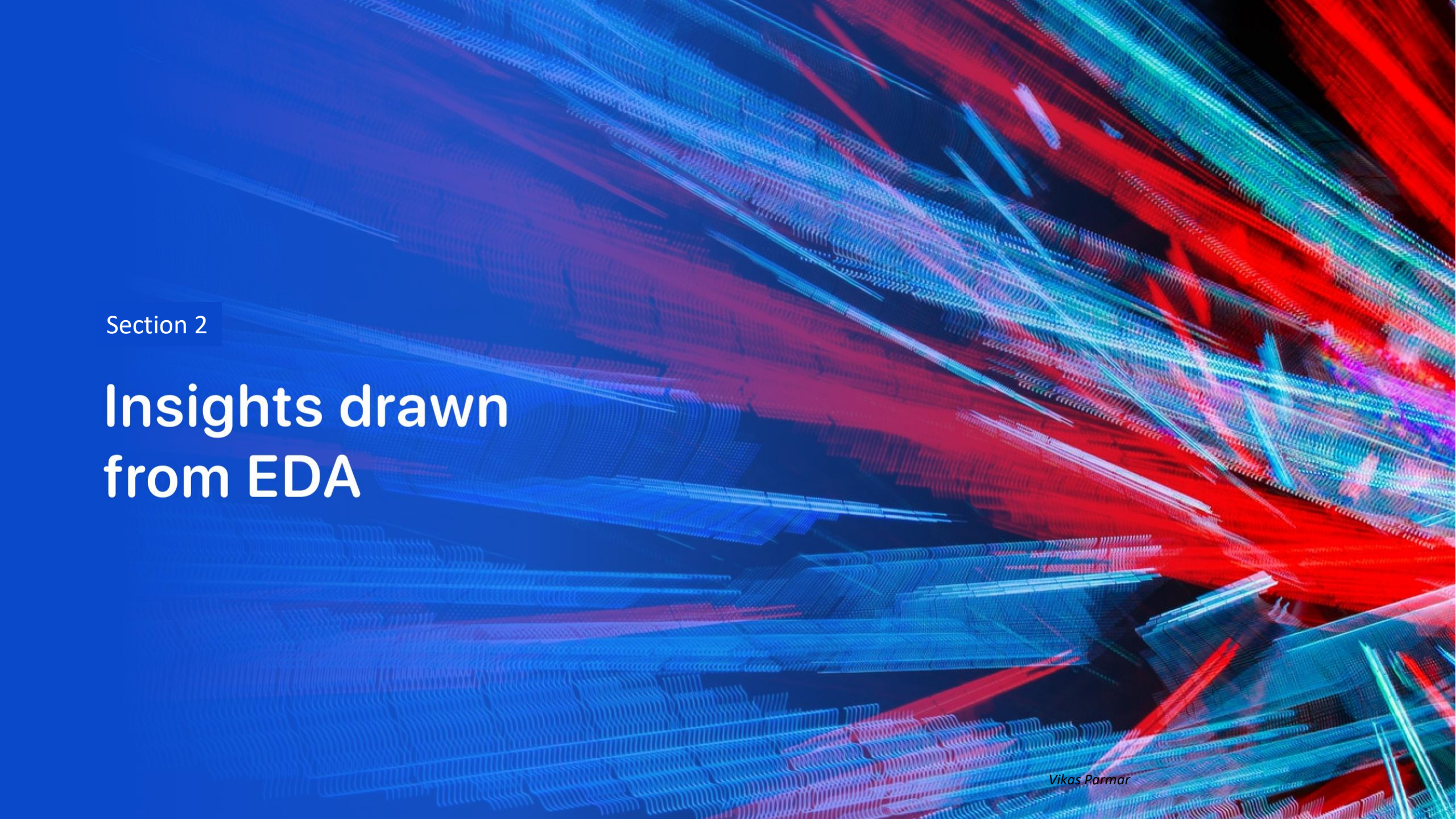
Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Predictive workflow:
Merge datasets ->
StandardScaler -> train/test
split -> GridSearchCV for
Logistic Regression, SVM,
Decision Tree, KNN ->
evaluate accuracy and
confusion matrix

Results

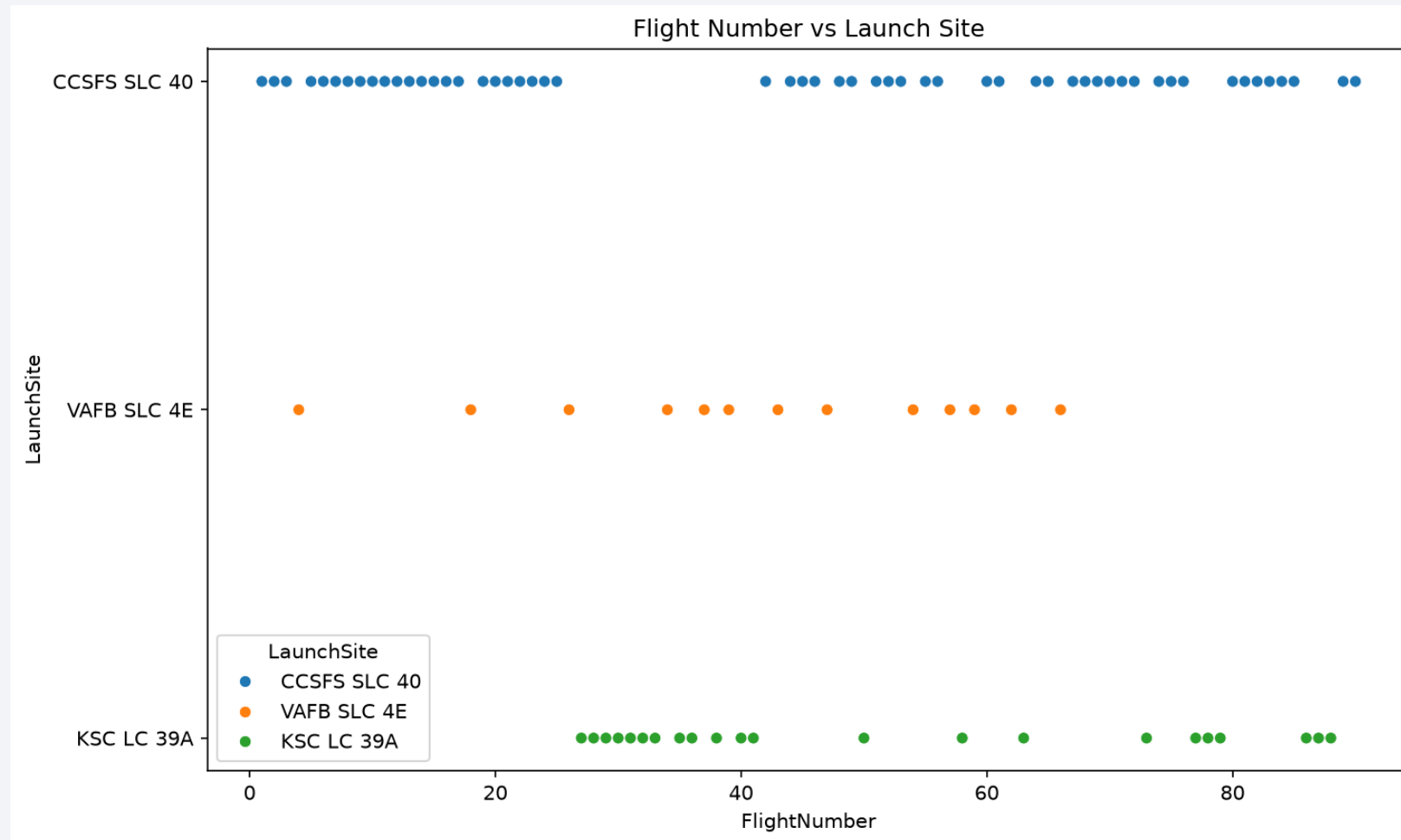
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



Section 2

Insights drawn from EDA

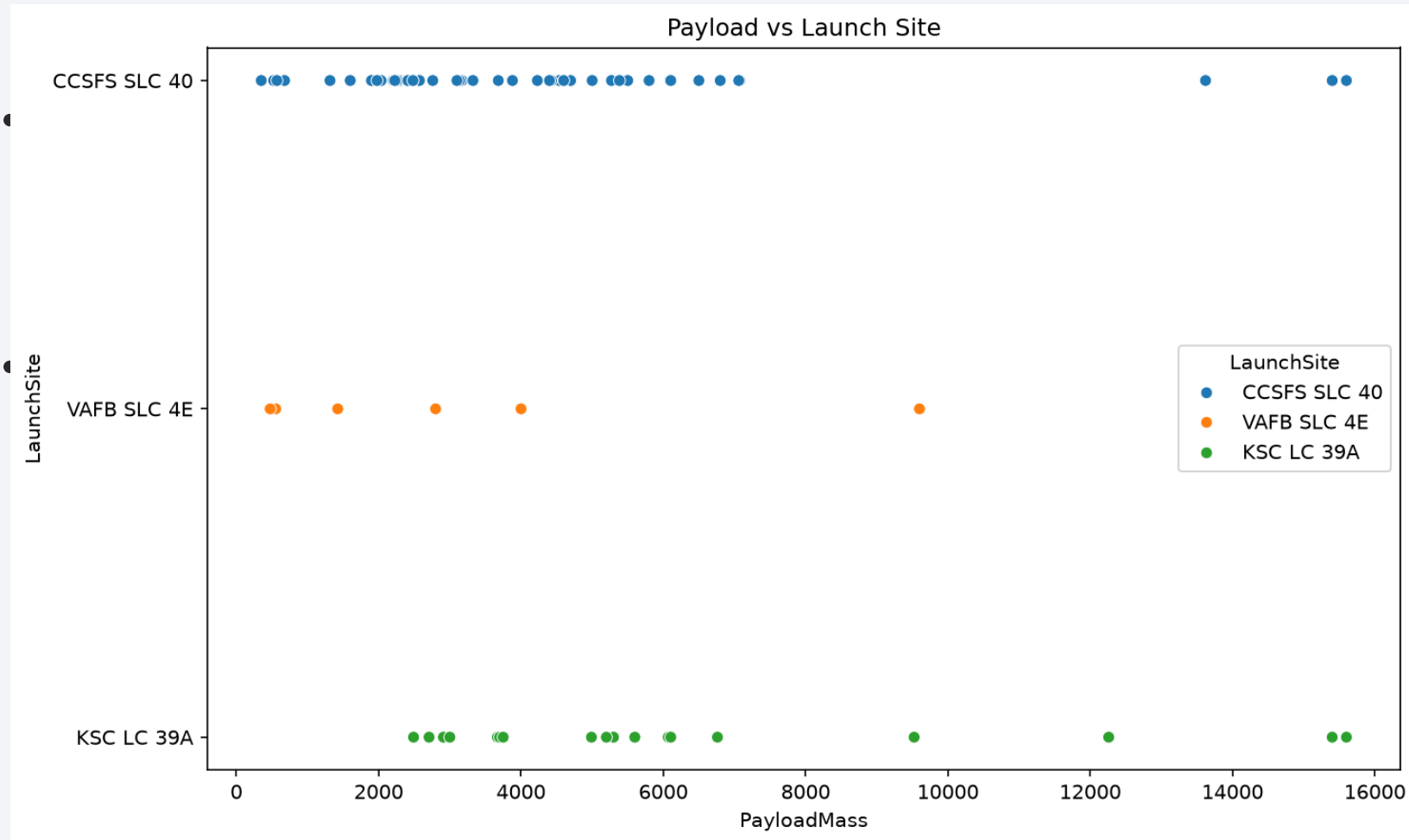
Flight Number vs. Launch Site



Flight number increases over
time at each launch site.

Vikas Parmar

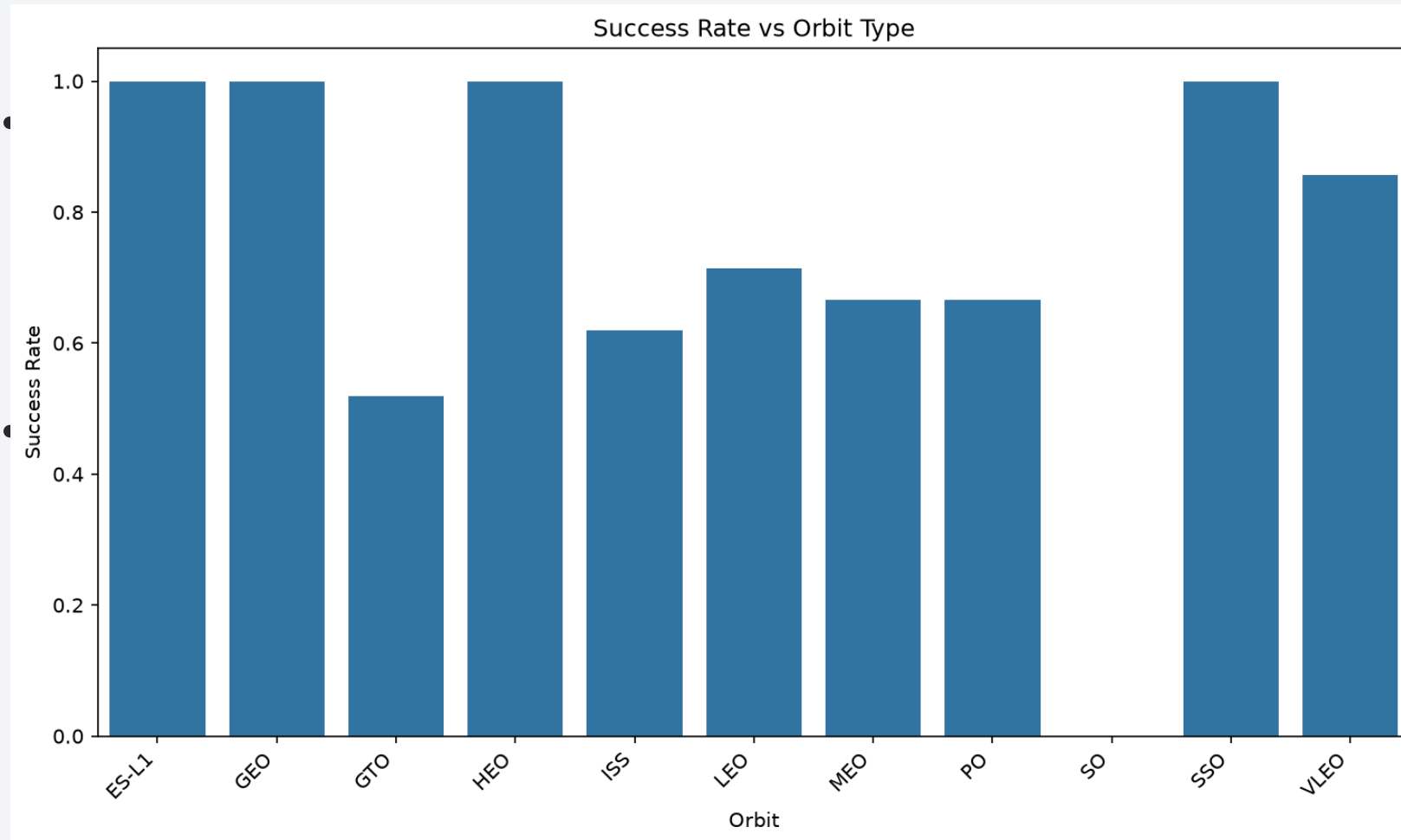
Payload vs. Launch Site



Payload distribution varies significantly by launch site.

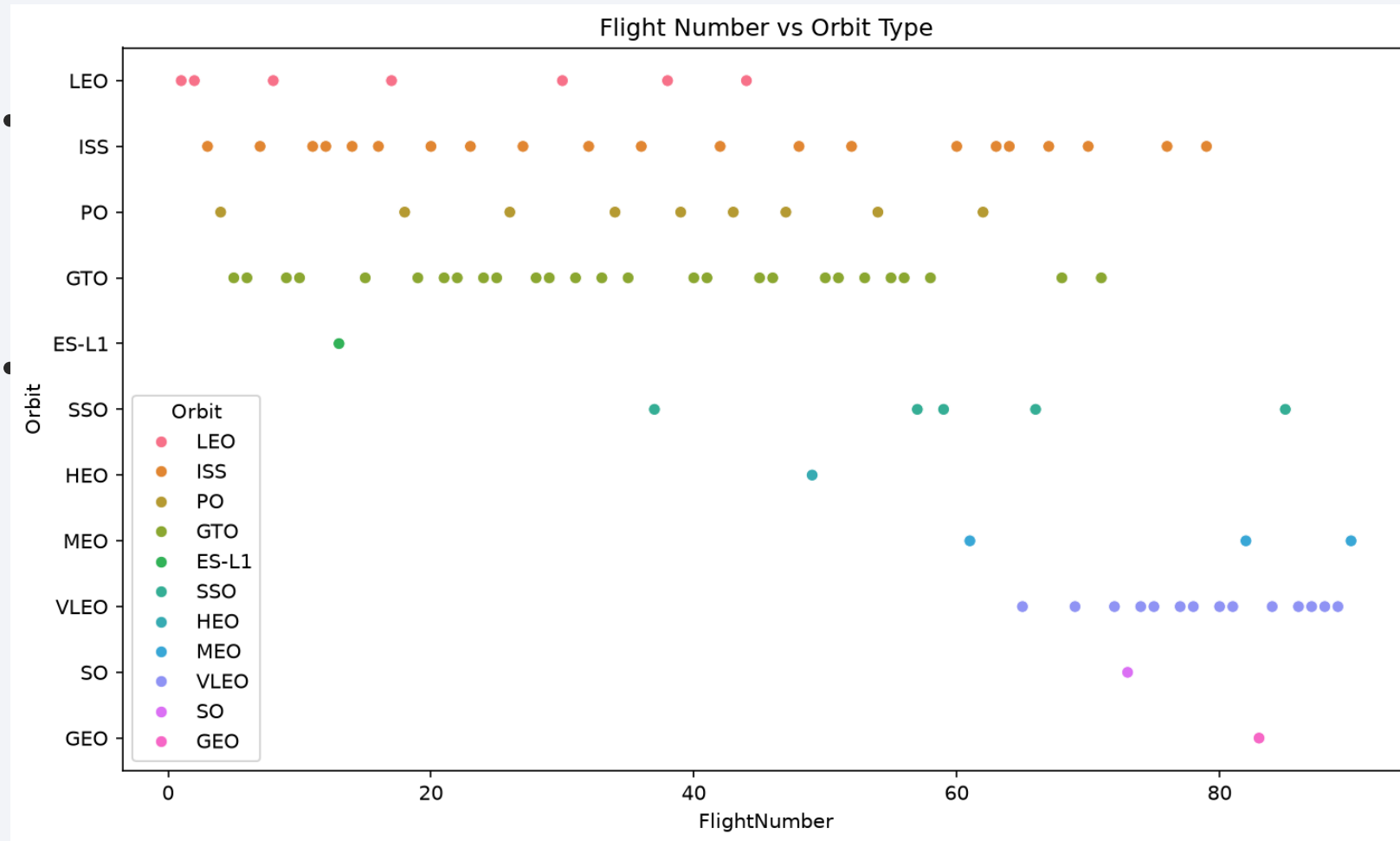
Vikas Parmar

Success Rate vs. Orbit Type



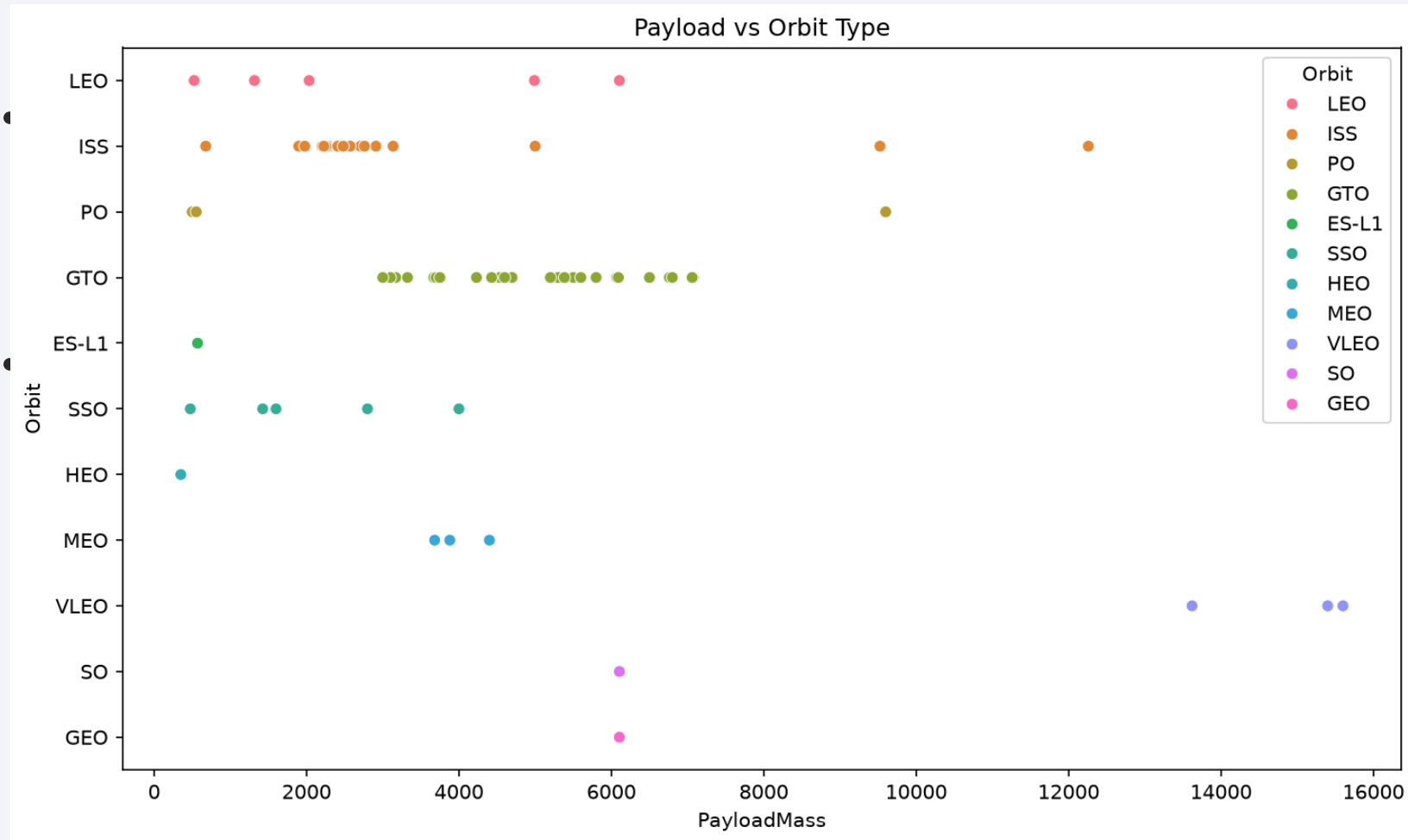
LEO and ISS orbits show
higher success rates than
GTO.

Flight Number vs. Orbit Type



Later flights show more diverse orbit assignments.

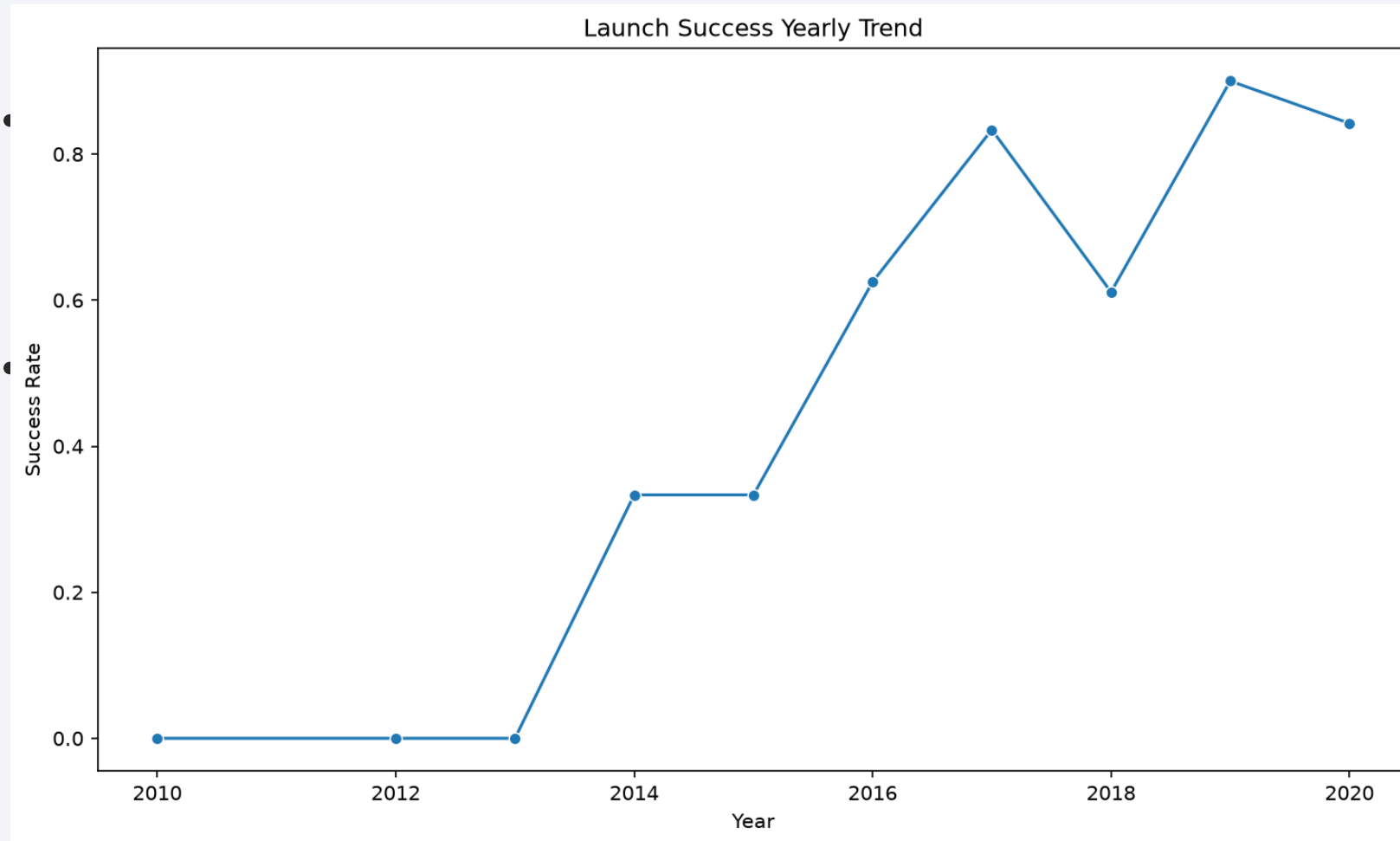
Payload vs. Orbit Type



Heavier payloads appear more often in GTO missions.

Vikas Parmar

Launch Success Yearly Trend



Landing success rate
improved sharply after 2014.

Vikas Parmar

All Launch Site Names

Sql 01 Launch Sites

LaunchSite
CCAFS SLC 40
VAFB SLC 4E
KSC LC 39A

Four unique launch sites
identified.

Launch Site Names Begin with 'CCA'

Sql 02 Cca Sites

LaunchSite
CCAFS SLC 40
CCAFS SLC 40
CCAFS SLC 40
CCAFS SLC 40
CCAFS SLC 40

CCAFS launch sites begin
with CCA prefix.

Total Payload Mass

Sql 03 Nasa Payload

SA

total_payload_mass

Total NASA CRS payload
mass calculated.

Average Payload Mass by F9 v1.1

Sql 04 Avg F9V11

version F9 v1.1

avg_payload_mass

Average payload for F9 v1.1
boosters.

Vikas Parmar

First Successful Ground Landing Date

Sql 05 First Ground Success

n ground pad

first_successful_ground_landing
2015-12-22 00:00:00

First successful RTLS landing
date.

Successful Drone Ship Landing with Payload between 4000 and 6000

Sql 06 Drone Ship Payload

BoosterVersion
Falcon 9
Falcon 9
Falcon 9
Falcon 9
Falcon 9
Falcon 9

ed on drone ship and 0

Boosters with drone ship landing and mid-range payload.

Total Number of Successful and Failure Mission Outcomes

Sql 07 Mission Outcomes

mission outcomes

success_count	failure_count
60	30

Successful vs failed mission
counts.

Boosters Carried Maximum Payload

Sql 08 Max Payload

maximum payload mass

BoosterVersion	PayloadMass
Falcon 9	15600.0
Falcon 9	15600.0
Falcon 9	15600.0

Boosters carrying maximum
payload mass.

2015 Launch Records

Sql 09 Failed 2015

Booster versions,

BoosterVersion	LaunchSite
Falcon 9	CCAFS SLC 40
Falcon 9	CCAFS SLC 40

- Present your query result with a short explanation here

Failed drone ship landings in
2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Sql 10 Rank Outcomes

Outcome	outcome_count
None None	9
True ASDS	5
False ASDS	4
True RTLS	3
True Ocean	3
None ASDS	2
False Ocean	2

one ship) or Success
-03-20, in

Ranked landing outcomes
between 2010 and 2017.

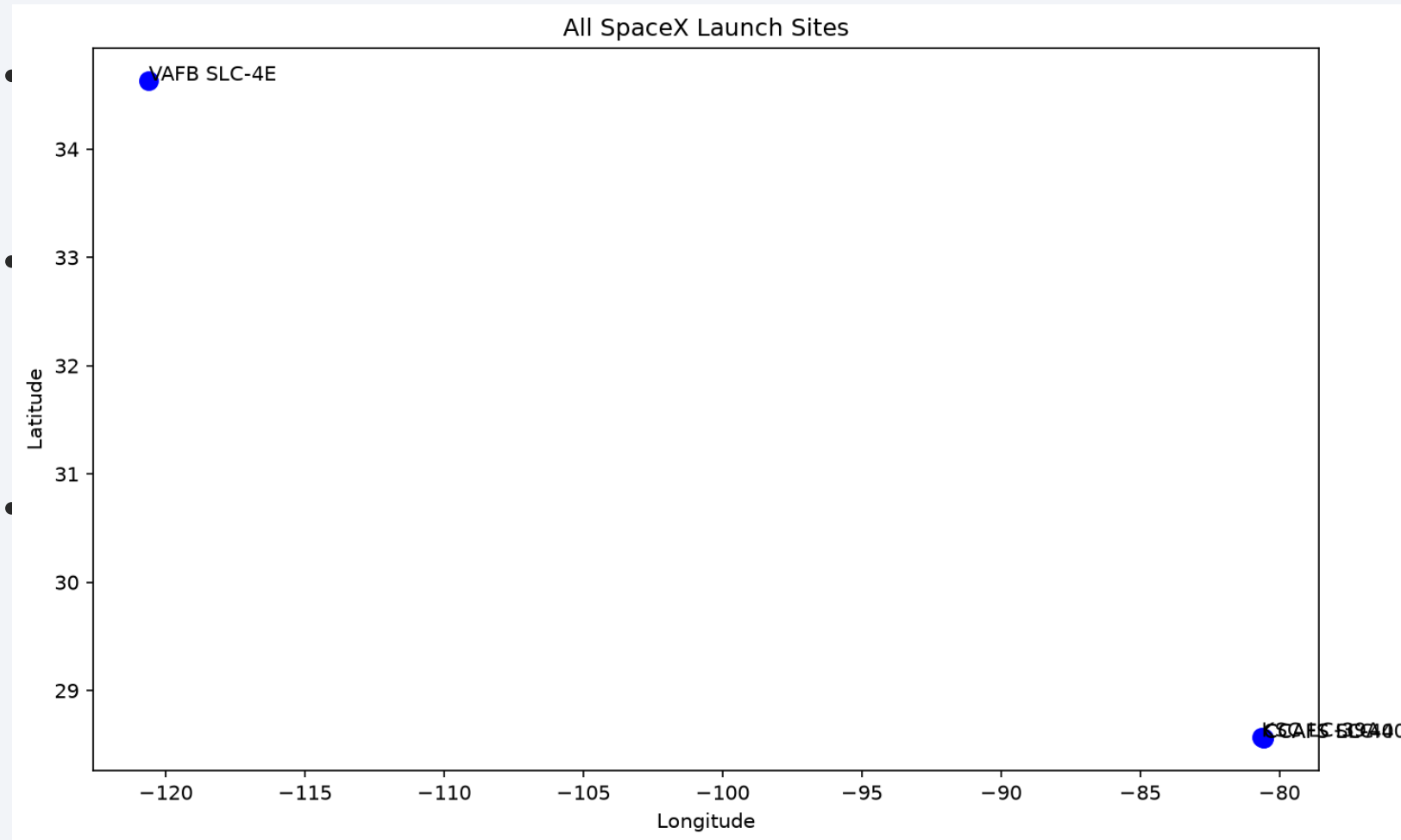
Vikas Parmar

The background of the slide is a high-quality satellite photograph of Earth taken from space. The image shows the dark blue of the night sky above the Earth's horizon. The Earth's surface is visible, with swirling white clouds and a dense network of yellow and orange lights representing city lights at night. The lights are concentrated in certain areas, forming a complex pattern across the visible landmasses.

Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>



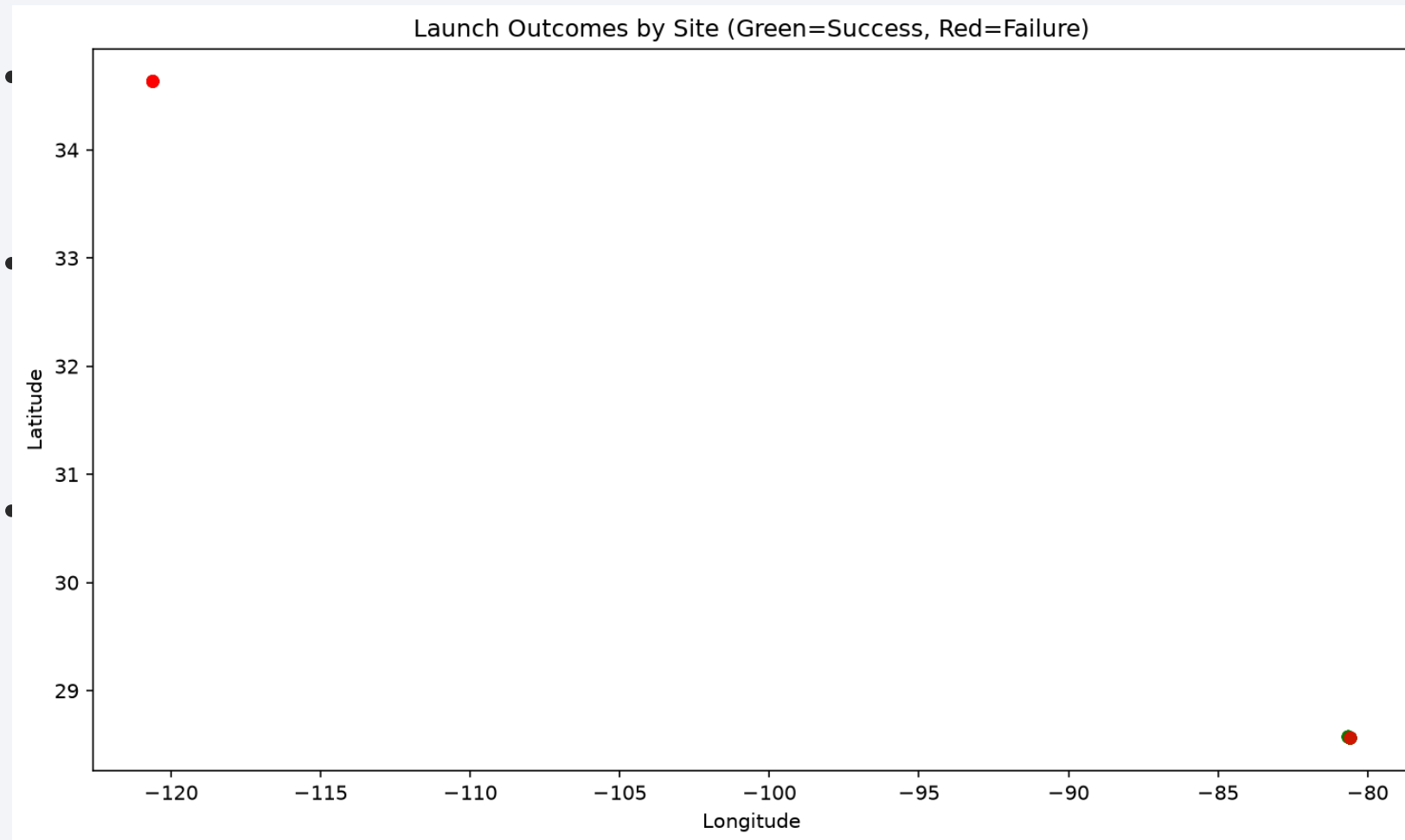
tle

ot to include

All four SpaceX launch sites
marked on the map.

Vikas Parmar

<Folium Map Screenshot 2>



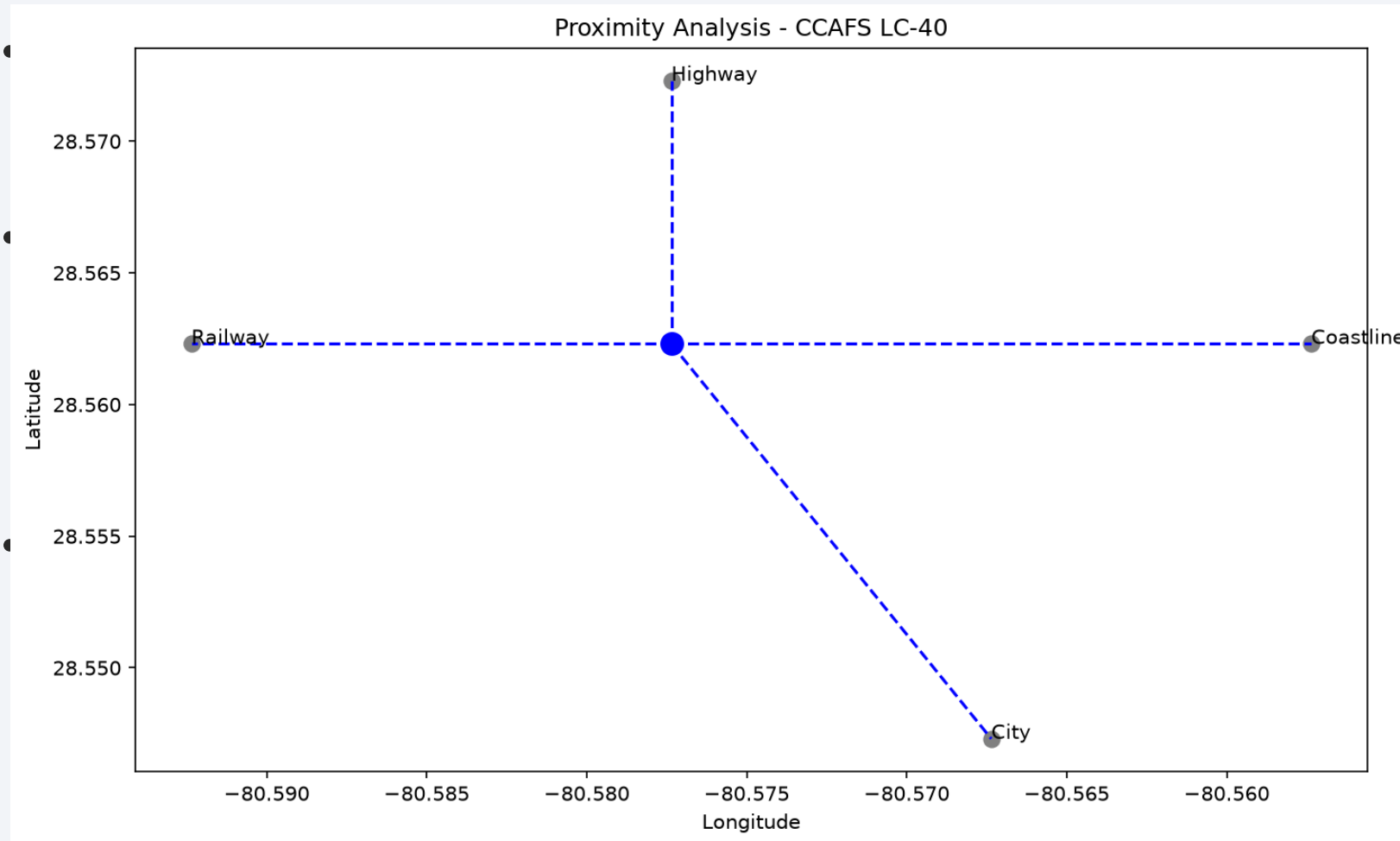
tle

the color-

Green markers = successful
landings; red = failures.

Vikas Parmar

<Folium Map Screenshot 3>

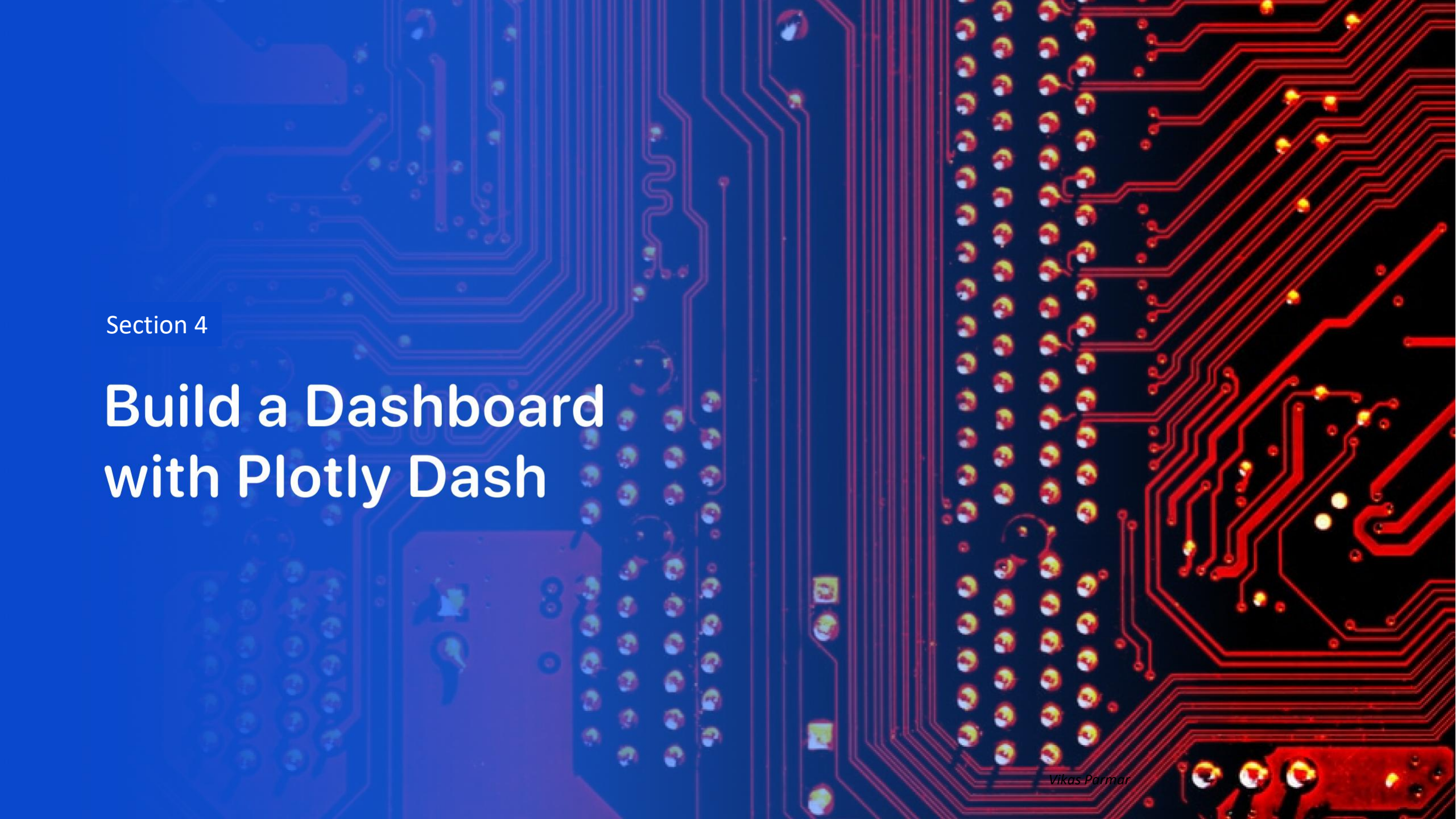


tle

λ

Proximity analysis shows
distance to key infrastructure.

Vikas Parmar



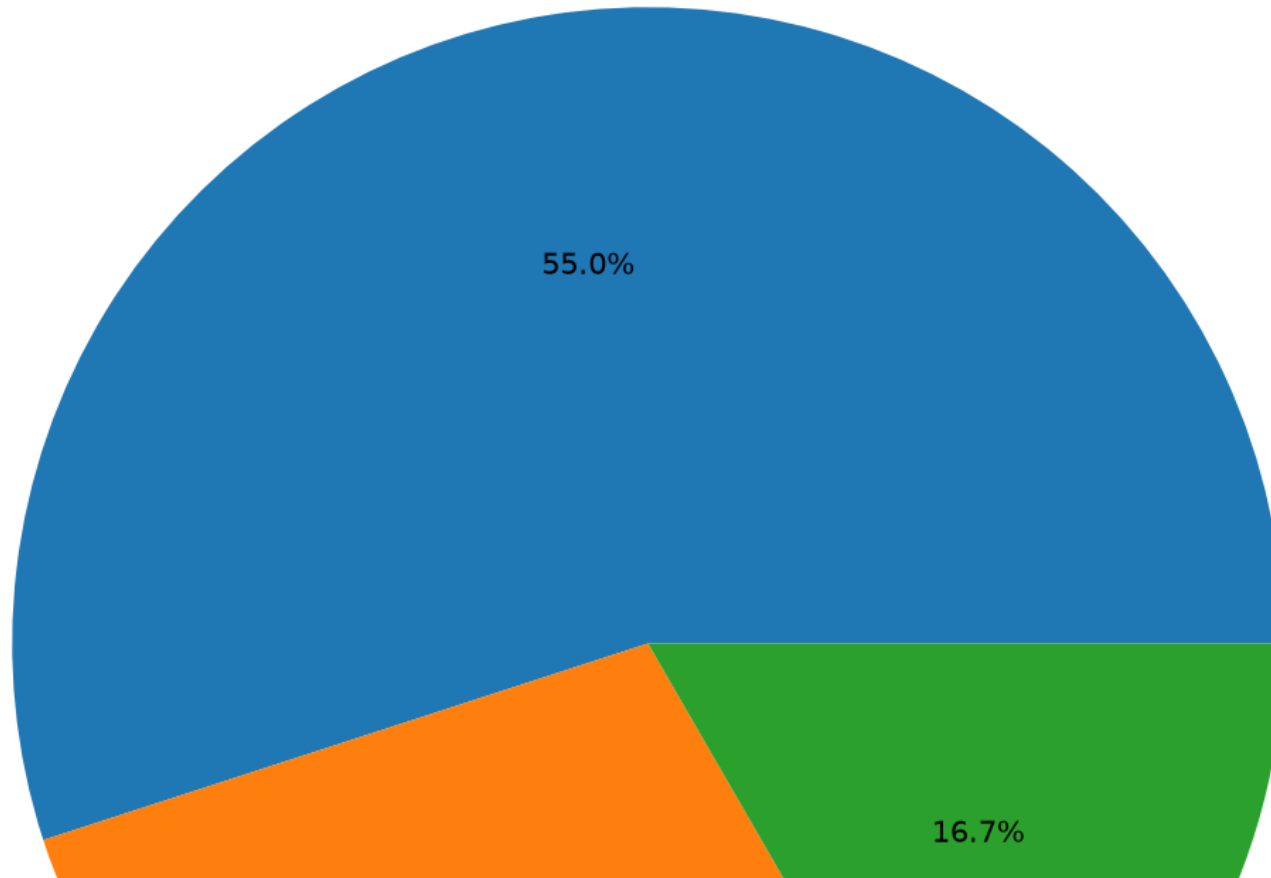
Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

Total Success Launches By Site

CCAFS SLC 40



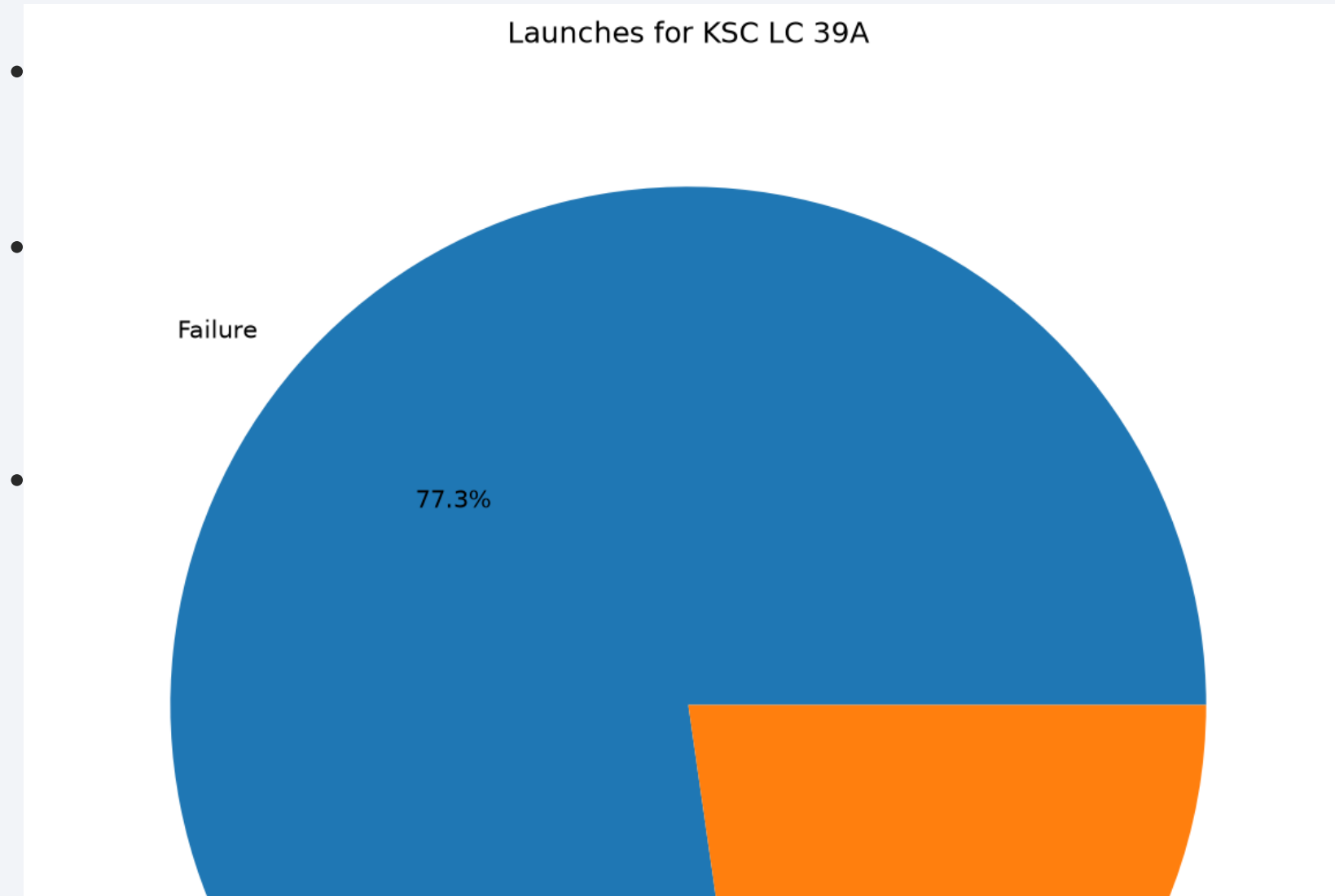
le

iechart

Success share by launch site
across all locations.

Vikas Parmar

<Dashboard Screenshot 2>



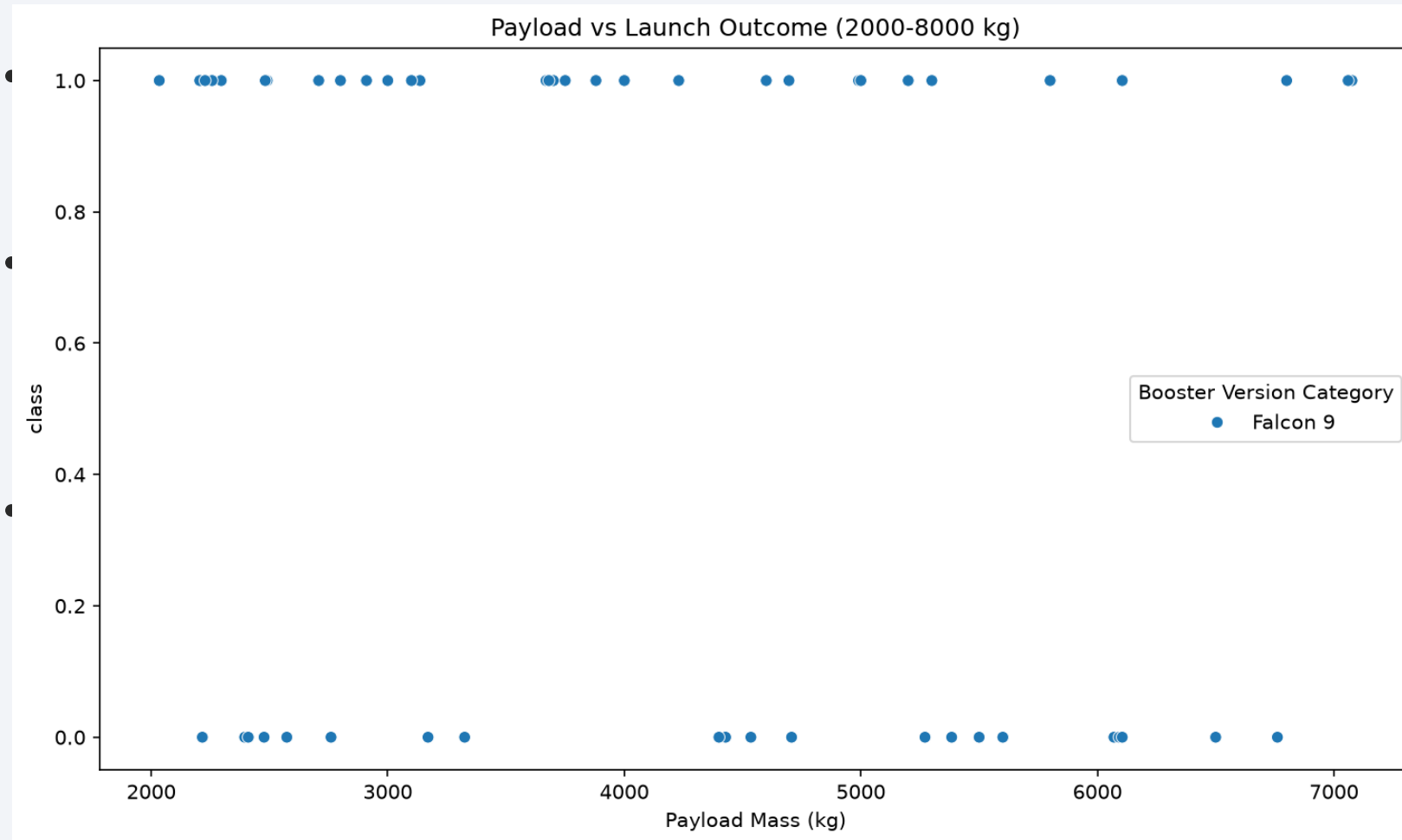
e

Best launch success

Site with highest success
ratio shown in pie chart.

Vikas Parmar

<Dashboard Screenshot 3>



le

or all sites, with

such as which
etc.

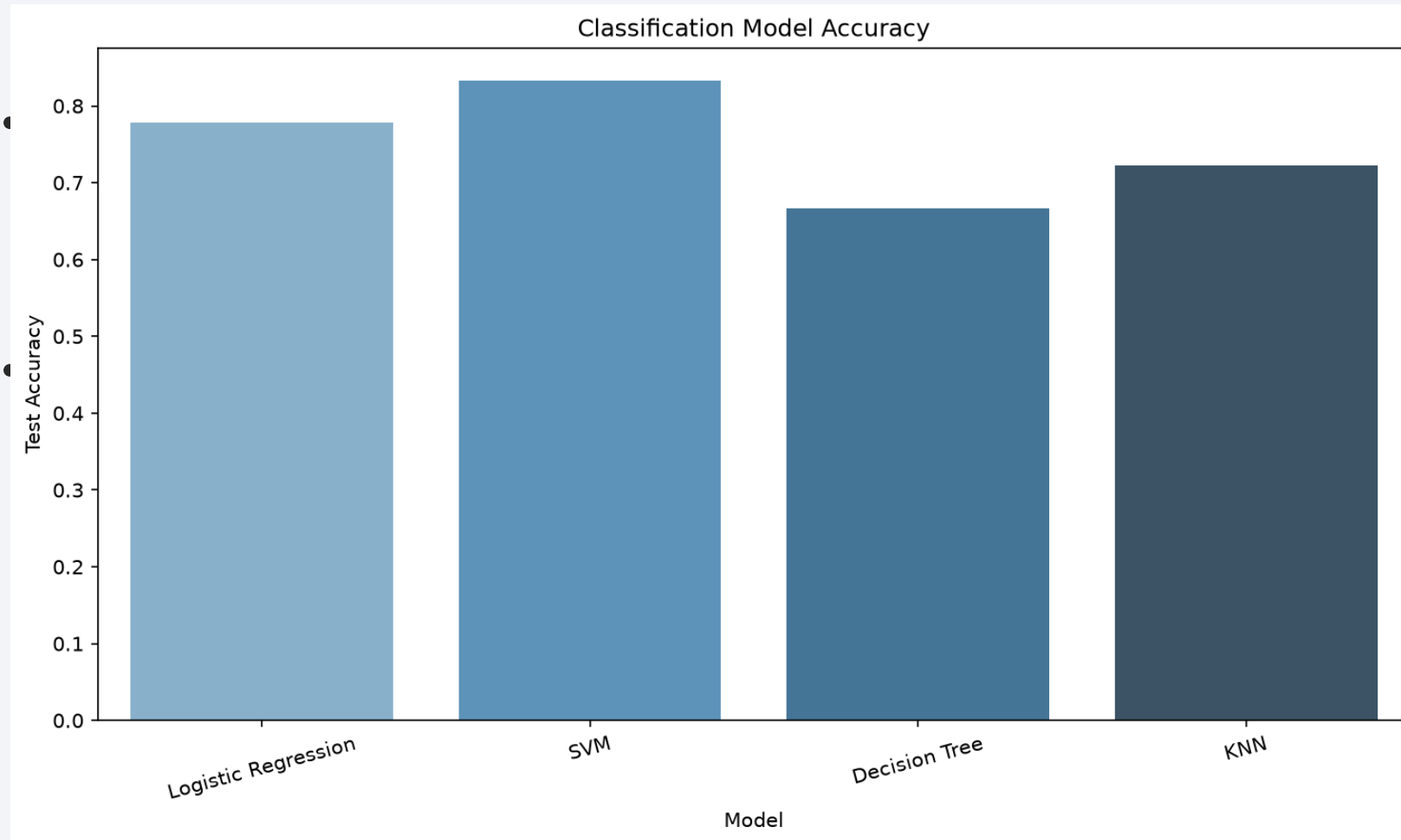
Payload vs outcome for
selected payload range.

Vikas Parmar

Section 5

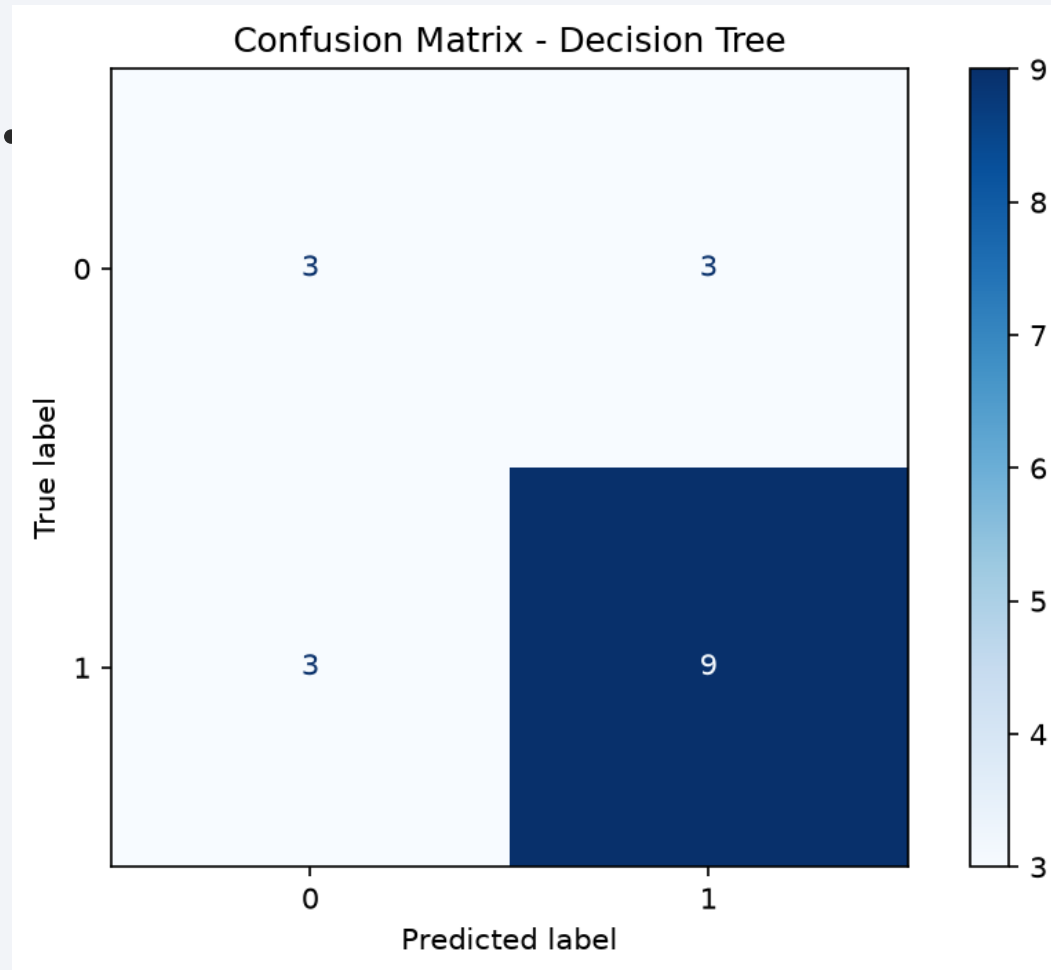
Predictive Analysis (Classification)

Classification Accuracy



Model comparison shows test accuracy for all four classifiers. Decision Tree has the highest GridSearchCV score.

Confusion Matrix



performing model with an

Confusion matrix for
Decision Tree. False positives
would overestimate reusable
first-stage availability and
understate launch cost.

Conclusions

- Point 1
- Point 2
- Point 3
- Point 4
- ...

1. Landing success improved significantly after 2014 as SpaceX gained experience.
2. Orbit type and payload mass strongly influence landing outcome.
3. Launch site selection affects both success rate and operational risk.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Author: Vikas Parmar

GitHub Repository:

<https://github.com/vikas-parmar/applied-data-science-capstone-ibm>

Vikas Parmar

Additional assets: notebooks,
SQL query outputs, charts

Thank you!



Vikas Parmar